Practical No:10	
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Batch	B3
Class	CSE
Academic Year & Semester	2026/27 & SEM-1
Date of Performance	22/10/2026
Title of Assignment:	Simple Sentiment Analysis of Tweets using Python
Objective	To understand the concept of sentiment analysis of social media text. To perform tweet sentiment classification using VADER polarity scores. To learn how to compute and interpret compound sentiment scores.
Theory:	**1. Sentiment Analysis** Sentiment analysis is a Natural Language Processing (NLP) technique used to determine whether a piece of text expresses positive, negative, or neutral emotion. It is widely used for analyzing tweets, reviews, feedback, and public opinion. **2. VADER (Valence Aware Dictionary and Sentiment Reasoner) – Short Explanation** VADER is a pre-trained, rule-based sentiment analysis tool specifically designed for social media text. It effectively handles: ● Slang (e.g., “awesome”, “terrible”) ● Emojis (😊 👿 😭) ● Emphasis (ALL CAPS, !!!, ???) ● Sentiment intensity words (“very good”, “extremely bad”) VADER produces four outputs:

	• **pos** → proportion of positive words • **neg** → proportion of negative words • **neu** → proportion of neutral content • **compound** → single score that summarizes overall sentiment **3. Compound Score – Short Explanation** The **compound score** is the most important output of VADER. It is a normalized score ranging from **–1 (most negative)** to **+1 (most positive)**. Using this score, sentiment is classified as: • **Positive** → compound ≥ +0.05 • **Negative** → compound ≤ –0.05 • **Neutral** → otherwise The compound score is preferred because it gives one final sentiment value while automatically considering intensity, punctuation, emojis, and emphasis.
Source Code/Algorithm/Flow Chart:	1. Load the dataset of tweets (CSV file). 2. Apply text preprocessing (remove URLs, mentions, emojis, symbols). 3. Initialize the VADER SentimentIntensityAnalyzer. 4. For each cleaned tweet, compute polarity scores. 5. Extract the compound score. 6. Classify sentiment as positive, negative, or neutral using thresholds.

	7. Store and display results. **8.** (Optional) Calculate accuracy if true labels exist.
Practice Programs:	

```
!pip3 install pandas nltk
import nltk
nltk.download('vader_lexicon')
import pandas as pd
import re
from nltk.sentiment import SentimentIntensityAnalyzer

# 1. Load the dataset
# The actual headers are in the second row (index 1)
of the CSV
# Based on the previous df.head() output,
pd.read_csv(header=1) resulted in 'Unnamed: X'
columns,
# and the actual headers ('id', 'tweetText') were
loaded as the first row of data (df.iloc[0]).
# We will load with header=1 as originally intended,
but then manually correct the column names.
df = pd.read_csv("tweets.csv", header=1)    # make
sure tweets.csv is in the same folder

# Correct column names: use the first row of the
loaded DataFrame as column headers
df.columns = df.iloc[0]
# Drop the row that was just used as headers
df = df[1:].reset_index(drop=True)

print("Sample data after correcting headers:")
print(df.head())

# 2. Basic preprocessing function
def clean_tweet(text):
    text = str(text)
    text = text.lower()
    text = re.sub(r"http\S+|www\S+|https\S+", "",
text)  # remove URLs
    text = re.sub(r"@\w+", "",
text)                      # remove mentions
    text = re.sub(r"#", "",
text)                          # remove hashtag symbol
    text = re.sub(r"[^a-zA-Z\s]", "",
text)              # keep only letters and spaces
    text = re.sub(r"\s+", " ",
text).strip()             # remove extra spaces
```

```
    return text

# Apply cleaning to the 'tweetText' column
df["clean_tweet"] =
df["tweetText"].apply(clean_tweet)

print("\nCleaned tweets:")
print(df[["tweetText", "clean_tweet"]].head())

# 3. Initialize VADER
sia = SentimentIntensityAnalyzer()

# 4. Function to get sentiment label from compound
score
def get_sentiment_label(score, pos_thresh=0.05,
neg_thresh=-0.05):
    if score >= pos_thresh:
        return "positive"
    elif score <= neg_thresh:
        return "negative"
    else:
        return "neutral"

# 5. Apply sentiment analysis
compound_scores = []
sentiment_labels = []

for t in df["clean_tweet"]:
    scores = sia.polarity_scores(t)
    #print(scores)
    compound = scores["compound"]
    compound_scores.append(compound)
    sentiment_labels.append(get_sentiment_label(compo
und))

df["compound_score"] = compound_scores
df["predicted_sentiment"] = sentiment_labels

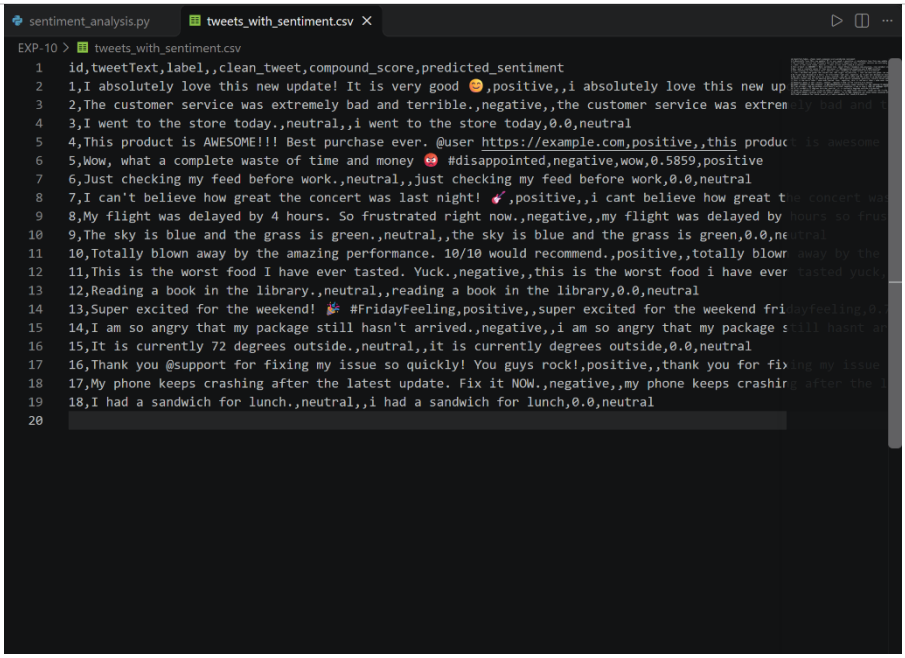
print("\nSentiment results:")
print(df[["tweetText", "clean_tweet",
"compound_score", "predicted_sentiment"]])

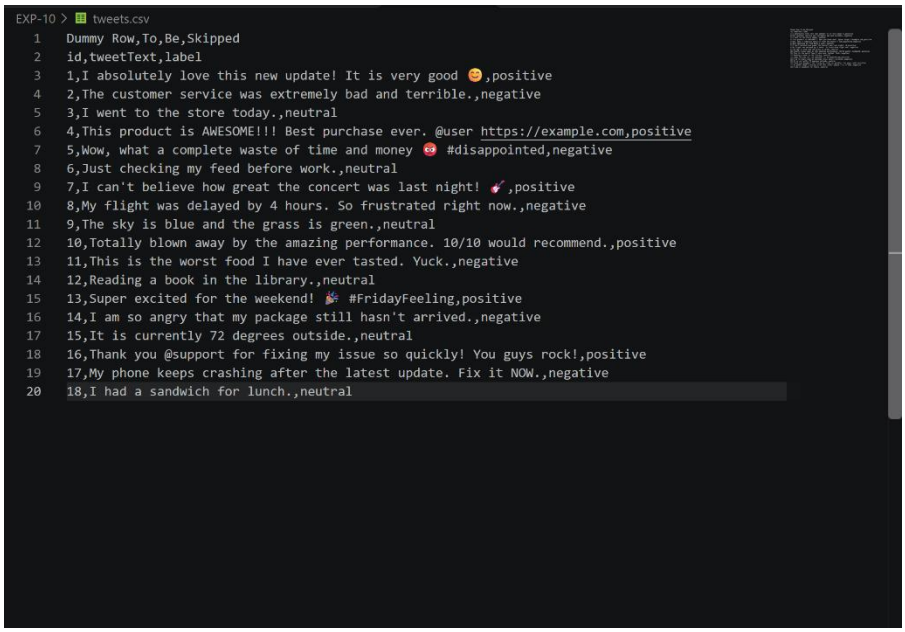
# 6. (Optional) Evaluate accuracy if true labels are
available
if "label" in df.columns:
    correct = (df["label"].str.lower() ==
df["predicted_sentiment"].str.lower()).sum()
    total = len(df)
    accuracy = correct / total * 100
    print(f"\nAccuracy: {accuracy:.2f}%
({correct}/{total})")

# 7. Save results to a new CSV
df.to_csv("tweets_with_sentiment.csv", index=False)
```

```
print("\nResults saved to tweets_with_sentiment.csv")
```

Output Screenshots





```
PS D:\SGA\EXP-10> pip install --upgrade nltk
PS D:\SGA\EXP-10> & C:\Users\Perin\AppData\Local\Programs\Python\Python315\python.exe d:/SGA/EXP-10/sentiment_analysis.py
[nltk_data] Downloading package vader_lexicon to
[nltk_data]     C:\Users\Perin\AppData\Roaming\nltk_data...
[nltk_data]   Package vader_lexicon is already up-to-date!
Sample data after correcting headers:
0 id  ...       NaN
0  1  ...       NaN
1  2  ...       NaN
2  3  ...       NaN
3  4  ...       NaN
4  5  ...  negative

[5 rows x 4 columns]

Cleaned tweets:
0                                          tweetText                                        clean_tweet
0  I absolutely love this new update! It is very ...  i absolutely love this new update it is very good
1  The customer service was extremely bad and ter...  the customer service was extremely bad and ter...
2                          I went to the store today.                          i went to the store today
3  This product is AWESOME!!! Best purchase ever...          this product is awesome best purchase ever
4                                                Wow                                                wow

Sentiment results:
0                                          tweetText  ... predicted_sentiment
0   I absolutely love this new update! It is very ...  ...           positive
1   The customer service was extremely bad and ter...  ...           negative
2                           I went to the store today.  ...            neutral
3   This product is AWESOME!!! Best purchase ever....  ...           positive
4                                                 Wow  ...           positive
5                Just checking my feed before work.  ...            neutral
6   I can't believe how great the concert was last...  ...           negative
7   My flight was delayed by 4 hours. So frustrate...  ...           negative
8            The sky is blue and the grass is green.  ...            neutral
9   Totally blown away by the amazing performance....  ...           positive
10   This is the worst food I have ever tasted. Yuck.  ...           negative
11                  Reading a book in the library.  ...            neutral
12    Super excited for the weekend! 🎉 #FridayFeeling  ...           positive
13  I am so angry that my package still hasn't arr...  ...           negative
14              It is currently 72 degrees outside.  ...            neutral
15  Thank you @support for fixing my issue so quic...  ...           positive
16  My phone keeps crashing after the latest updat...  ...            neutral
17                        I had a sandwich for lunch.  ...            neutral

[18 rows x 4 columns]

Accuracy: 83.33% (15/18)

Results saved to tweets_with_sentiment.csv
PS D:\SGA\EXP-10>
```

Conclusion

In this practical, we successfully performed sentiment analysis on tweets using Python and VADER. The VADER SentimentIntensityAnalyzer efficiently classified tweets as positive, negative, or neutral based on the compound score. This demonstrates how lexicon-based sentiment tools can analyze social media text quickly and accurately.